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Section:- 4A

Course:- PRP

Q#01

The probability that the reason of theft is money for drug is

$$P = 75/100 \\ = 0.75$$

Let x represents the number of money thefts to buy drugs among the next $x=5$ theft cases. The probability distribution of random variable x is

$$P(X=x) = \binom{5}{x} (0.75)^x (1-0.75)^{5-x} \\ = \binom{5}{x} (0.75)^x (0.25)^{5-x}, x=0,1,\dots$$

(a) We need to find $P(X=2)$
we have

$$P(X=2) = \binom{5}{2} (0.75)^2 (0.25)^{5-2} \\ = \frac{5!}{2!3!} (0.5625)(0.0156) \\ = 10(0.5625)(0.0156) \\ = 0.0879$$

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(b) We need to find $P(X \leq 3)$

$$P(X \leq 3) = P(X=1) + P(X=2) + P(X=3)$$
$$= \binom{5}{1} (0.75)(0.25)^4 + \binom{5}{2} (0.75)^2 (0.25)^3 + \binom{5}{3} (0.75)^3 (0.25)$$
$$= 0.3672$$

Q#02

(a) We need to find the probability that all the 4 will fire.

We have a random sample of $x=4$ taken from $N=10$ items of which $k=3$ and $N-k=10-3=7$ are labelled failure. the probability distribution of hypergeometric random variable x is

$$h(x; N, n, k) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}} \quad \text{--- (1)}$$

We need to find $P(X=0) = h(0; 10, 4, 3)$

According to eq. (1), we have

$$h(x; N, n, k) = h(0; 10, 4, 3) = \frac{\binom{3}{0} \binom{10-3}{4-0}}{\binom{10}{4}}$$

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$$= \frac{\binom{3}{0} \binom{7}{4}}{\binom{10}{4}}$$

$$= \frac{(1)(35)}{210} = \frac{1}{6}$$

(b) We need to find $P(0 \leq x \leq 2)$
we have

$$\begin{aligned} P(0 \leq x \leq 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= h(0; 10, 4, 3) + h(1; 10, 4, 3) \\ &\quad + h(2; 10, 4, 3) \quad \text{--- ①} \end{aligned}$$

According to (question)

$$h(0; 10, 4, 3) = \frac{1}{6}$$

$$h(1; 10, 4, 3) = \frac{\binom{3}{1} \binom{10-3}{4-1}}{\binom{10}{x}}$$

$$= \frac{\binom{3}{1} \binom{7}{3}}{\binom{10}{x}}$$

$$= \frac{1}{2}$$

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$$x(2; 10, 4, 3) = \frac{\binom{3}{2} \binom{10-3}{4-2}}{\binom{10}{4}}$$

$$= \frac{\binom{3}{2} \binom{7}{2}}{\binom{10}{4}}$$

$$= \frac{3}{10}$$

Putting these values in eq - ①

$$P(0 \leq x \leq 2) = x(0; 10, 4, 3) + x(1; 10, 4, 3) + x(2; 10, 4, 3)$$

$$= \frac{1}{6} + \frac{1}{2} + \frac{3}{10}$$

$$= \frac{29}{30}$$

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Q#03

Total number of cases $N=10,000/-$

Number of underfilled cans $k=300$

No. of cases selected ~~cans~~ as sample $x=30$

the probability mass function of x is

$$P(X=x) = \frac{\binom{k}{x} \binom{N-k}{x-x}}{\binom{N}{x}}$$

$$= \frac{\binom{300}{x} \binom{10,000-300}{30-x}}{\binom{10,000}{30}}$$

$$= \frac{\binom{300}{x} \binom{9700}{30-x}}{\binom{10,000}{30}}, \quad 0 \leq x \leq 30$$

Now finding $P(X \geq 1)$

$$P(X \geq 1) = 1 - P(X < 1)$$

$$= 1 - P(X=0)$$

$$= 1 - \frac{\binom{300}{0} \binom{9700}{300-0}}{\binom{10000}{30}}$$

$$= 1 - 0.4005$$

$$= 0.5995$$

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Q#4

Probability of ~~trial~~ a success in each trial is $p = \frac{2}{3}$

Probability of a failure in each trial is
 $q = 1 - p$
 $q = 1/3$

- (a) Because trials are independent X has a negative distribution with parameters $k=1$ and $p=2/3$

$$P(X=x) = b(x; k, p)$$

$$= \binom{x-1}{k-1} p^k q^{x-k}$$

$$= \binom{x-1}{1-1} \left(\frac{2}{3}\right)^1 \left(\frac{1}{3}\right)^{x-1}$$

$$= \binom{2}{3} \left(\frac{1}{3}\right)^{x-1}, x=1, 2, 3 \quad \text{--- ①}$$

We need to find $P(X=5)$
using eq. ①

$$P(X=5) = \binom{2}{3} \left(\frac{1}{3}\right)^{5-1}$$

$$= 0.0082$$

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- (b) Because trials are independent, X has a negative binomial distribution with parameters $k=3$ and $p=2/3$

$$P(X=x) = \binom{x-1}{k-1} p^k q^{x-k}$$

$$= \binom{x-1}{3-1} \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^{x-3}$$

$$= \binom{4}{2} \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^2$$

$$= 0.1975$$

Q # 5

- (a) Average number of emergency patients per hour is $\lambda=5$

Since, the given interval of time is 1 hour, $t=1$
Here, X has a poisson distribution with parameters

$$\lambda t = (5)(1) = 5$$

Probability mass function of X is

$$P(X=x) = P(x; \lambda t), \text{ where } x=0,1,2,3, \\ = e^{-\lambda t} \frac{(\lambda t)^x}{x!}$$

Lets create the probability that in a given hour the staff cannot accomodate the patient traffic

$$P(X > 10) = 1 - P(X \leq 10)$$

$$= 1 - \sum_{x=0}^{10} P(x; 5)$$

$$= 1 - 0.986$$

$$= 0.0137$$

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(b) Average number of emergency patients per hour is $\lambda = 5$

since the given intervals of time is 3 hour, $t = 3$

Hence, X has a poisson distribution with parameters

$$\lambda t = 5 \times 3 = 15$$

Probability mass function of X is

$$P(X = x) = p(x; \lambda t), \text{ where } x = 0, 1, 2, 3$$
$$= e^{-\lambda t} \frac{(\lambda t)^x}{x!}$$

Now, let's find the probability that more than 20 emergencies arrive during a 3 hour shift

$$P(X > 20) = 1 - P(X \leq 20)$$
$$= 1 - \sum_{x=0}^{20} P(X; 15)$$

$$= 1 - 0.9170$$

$$= 0.0880$$

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Q#06

- (a) The normal distribution with $\mu=30$ and $\sigma=6$
Let's calculate the normal area curve to the right of $x=17$.

$$Z = \frac{x - \mu}{\sigma}$$

$$= \frac{17 - 30}{6} = -2.17$$

Hence,

$$P(X > 17) = P(Z > -2.17)$$

$$= 1 - P(Z \leq -2.17)$$

$$= 1 - 0.0150$$

$$= 0.9850$$

- (b) Let's determine the normal area curve to the left at $x=22$

$$Z = \frac{x - \mu}{\sigma} = \frac{22 - 30}{6}$$

$$Z = -1.33$$

Therefore,

$$P(X < 22) = P(Z < -1.33)$$

$$= 0.0918$$

- (c) Let's determine the normal area curve between $x=32$ and $x=41$

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$$Z_1 = \frac{x_1 - \mu}{\sigma} = \frac{32 - 30}{6} = 0.33$$

$$Z_2 = \frac{x_2 - \mu}{\sigma} = \frac{41 - 30}{6} = 1.83$$

Hence,

$$\begin{aligned} P(32 < X < 41) &= P(X < 41) - P(X < 32) \\ &= P(Z_2 < 1.83) - P(Z_1 < 0.33) \\ &= 0.9664 - 0.6293 \\ &= 0.3371 \end{aligned}$$

(d) Let's find the value of x that has 80% of the normal curve area to the left.

From the normal tables, we get the values of Z that an area of 0.80 to the left of 0.842

Therefore,

$$Z = \frac{x - \mu}{\sigma}$$

$$x = \sigma Z + \mu$$

$$x = 6(0.842) + \mu$$

$$x = 6(0.842) + 30$$

$$x = 35.05$$

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(e) Let's find the two values of x that contains the middle of 75% of the normal curve area

$$P(-z \leq Z \leq z) = 0.75$$

$$P(Z \leq z) - P(Z \leq -z) = 0.75$$

$$2P(Z \leq z) - 1 = 0.75$$

$$2P(Z \leq z) = 1.75$$

$$P(Z \leq z) = 0.875$$

Therefore, the values of z that area of 0.875 to the right in $z = -1.15$ and left is $z_2 = 1.15$,

$$z_1 = \frac{x_1 - \mu}{\sigma}$$

$$\begin{aligned} x_1 &= \sigma z_1 + \mu \\ &= 6 \cdot (-1.15) + 30 \\ &= 23.1 \end{aligned}$$

$$z_2 = \frac{x_2 - \mu}{\sigma}$$

$$\begin{aligned} x_2 &= \sigma z_2 + \mu \\ x_2 &= 6(1.15) + 30 \\ x_2 &= 36.9 \end{aligned}$$

Q #07

Let random variable X represent the height of students that follows normal distribution with mean 174.5 and the standard deviation of 6.9 of 6.9 cm

- (a) Let's find how many of these students would you expect to have height less than 160.0 cm

$$\begin{aligned}
 P(X \leq 160.0) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{160 - 174.5}{6.9}\right) \\
 &= P\left(Z \leq \frac{-14.5}{6.9}\right) \\
 &= P(Z \leq -2.10) \\
 &= 0.0179
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 E(X) &= np \\
 &= (1000)(0.0179) \\
 &= 17.9 \approx 18
 \end{aligned}$$

- (b) Lets calculate how many students of these type would you expect to have heights between 171.5 and 182.0 cm inclusive

$$\begin{aligned}
 P(171.5 \leq X \leq 182.0) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{182 - 174.5}{6.9}\right) \\
 &= P\left(Z \leq \frac{7.5}{6.9}\right) = P(Z \leq 1.0870) \\
 &= 0.8621
 \end{aligned}$$

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$$\begin{aligned}P(X \leq 171.5) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{171.5 - 174.5}{6.9}\right) \\&= P\left(Z \leq \frac{-3}{6.9}\right) = P(Z \leq -0.4348) \\&= 0.3336\end{aligned}$$

$$\begin{aligned}P(171.5 \leq X \leq 182) &= P(X \leq 182) - P(X \leq 171.5) \\&= 0.8621 - 0.3336 \\&= 0.5285\end{aligned}$$

Therefore,

$$\begin{aligned}E(X) &= np \\&= (1000)(0.5285) \\&= 528.5 \approx 529\end{aligned}$$

(c) Lets find how many of these students would you expect to have heights equal to 175.0 cm

$$P(X \leq 175.5) = P\left(\frac{X - \mu}{\sigma} \leq \frac{175.5 - 174.5}{6.9}\right)$$

$$= P(Z \leq 0.14) = 0.5557$$

$$P(X \leq 174.5) = P\left(\frac{X - \mu}{\sigma} \leq \frac{174.5 - 174.5}{6.9}\right)$$

$$= P(Z \leq 0) = 0.5$$

$$P(175 - 0.5 \leq X \leq 175 + 0.5) = P(174.5 \leq X \leq 175.5)$$

$$= P(X \leq 175.5) - P(X \leq 174.5)$$

$$= 0.5557 - 0.5$$

$$= 0.0557$$

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Therefore,

$$\begin{aligned} E(X) &= np \\ &= (1000)(0.557) \\ &= 55.7 \approx 56 \end{aligned}$$

(d) Now, Let's find how many of these students would you expect to have heights greater than or equal to 180 cm

$$\begin{aligned} P(X \geq 180) &= 1 - P(X \leq 188.0) \\ &= 1 - P\left(\frac{X - \mu}{\sigma} \leq \frac{188.0 - 174.5}{6.9}\right) \\ &= 1 - P(Z \leq 1.96) \\ &= 1 - 0.9750 \\ &= 0.025 \end{aligned}$$

$$\begin{aligned} E(X) &= np \\ &= (1000)(0.025) \\ &= 25 \end{aligned}$$

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Q#08

Let random variable X represents the daily consumption of electric power, in millions of kilowatt-hour.

X has a gamma distributions with mean $\mu=6$ & variance of $\sigma^2=12$

The mean of gamma distribution is

$$\mu = \alpha\beta$$

$$\alpha\beta = 6 \quad \text{--- (1)}$$

the variance of gamma distribution is

$$\sigma^2 = \alpha\beta^2$$

$$\alpha\beta^2 = 12 \quad \text{--- (2)}$$

(a)

Let's find the values of A and B

using the eq. - (1), we get $\alpha = 6/\beta$. Now let's substitute this value in eq. - (2)

$$\alpha\beta^2 = 12$$

$$\frac{6}{\beta} \beta^2 = 12$$

$$6\beta = 12$$

$$\beta = 2$$

$$\alpha = \frac{6}{\beta}$$

$$\alpha = \frac{6}{2}$$

$$\alpha = 3$$

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(b) The probability distribution of gamma distribution

$$f(x; 3, 2) = \begin{cases} \frac{1}{(2)^3 (3)} x^{3-1} e^{-x/2}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Now, let's calculate the probability

$$P(X > 12) = 1 - P(X \leq 12)$$

$$= 1 - \int_0^{12} \frac{1}{16} x^2 e^{-x/2} dx$$

$$= 1 - \frac{1}{16} \int_0^{12} x^2 e^{-x/2} dx$$

Integrating by parts

$$= 1 - \frac{1}{16} \left[x^2 \int e^{-x/2} dx - \int \left(\frac{d}{dx} (x^2) \right) \left(\int e^{-x/2} dx \right) dx \right]_0^{12}$$

$$= 1 - \frac{1}{16} \left[x^2 (-2e^{-x/2}) - \int 2x (-2e^{-x/2}) dx \right]_0^{12}$$

$$= 1 - \frac{1}{16} \left[-2x^2 e^{-x/2} + 4 \int x e^{-x/2} dx \right]_0^{12}$$

$$= 1 - \left(-\frac{1}{8} \left[x^2 e^{-x/2} - 2 \int x e^{-x/2} dx \right]_0^{12} \right)$$

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Integrating by parts again

$$\begin{aligned} &= 1 + \frac{1}{8} \left[x^2 e^{-x/2} - 2 \left(x \int e^{-x/2} dx - \int \frac{d}{dx} (x) \int e^{-x/2} dx \right) \right]_0^{12} \\ &= 1 + \frac{1}{8} \left[x^2 e^{-x/2} - 2 \left(x(-2e^{-x/2}) - \int (-2e^{-x/2}) dx \right) \right]_0^{12} \\ &= 1 + \frac{1}{8} \left[x^2 e^{-x/2} - 2(-2xe^{-x/2} + 2(-2e^{-x/2})) \right]_0^{12} \end{aligned}$$

$$\begin{aligned} &= 1 + \frac{1}{8} \left[x^2 e^{-x/2} + 4xe^{-x/2} + 8e^{-x/2} \right]_0^{12} \\ &= 1 + \frac{1}{8} \left[(12)^2 e^{-12/2} + 4(12)e^{-12/2} + 8e^{-12/2} \right] - \end{aligned}$$

$$\begin{aligned} &\quad \left[(0)^2 e^{-0/2} + 4(0)e^{-0/2} + 8e^{-0/2} \right] \\ &= 1 + \frac{1}{8} [200e^{-6} - 8] \end{aligned}$$

$$= 1 - 1 + 25e^{-6}$$

$$= 0.0620$$

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Q #09

Let random variable x represent the number of years that the switch survived. x follows an exponential distribution with a mean $B=2$. The probability density function of exponential distribution is

$$f(x) = \frac{1}{B} e^{-(x/B)} \quad x > 0$$

the probability distribution of the exponential distribution is

$$\begin{aligned} P(X \leq x) &= 1 - e^{-(x/B)} \\ &= 1 - e^{-x/2} \quad x > 0 \end{aligned}$$

First, let's find the probability the number of switches fail during the first year.

$$\begin{aligned} P(X < 1) &= 1 - e^{-x/2} \\ &= 1 - e^{-1/2} \\ &= 1 - e^{-0.5} \\ &= 0.3935 \end{aligned}$$

The total number of electrical switches $x=100$ switch

We considered it success if $\sim$ fail during the first year.

The probability of success is $p=0.3935$

Hence, the probability of failure is $q=1-p=1-0.3935$

$$q = 0.6065$$

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Let, random variable Y represents the number of switches failed during the first year.

Therefore, X has binomial distribution with parameter $n=100$ and $P=0.3935$.

Let's find the probability that the most 30 fail during the first year.

$$\begin{aligned} P(X \leq 30) &= P(X \leq 30.5) \\ &= P\left(\frac{X - np}{\sqrt{npq}} \leq \frac{30.5 - np}{\sqrt{npq}}\right) \\ &= P\left(Z \leq \frac{30.5 - (100 \times 0.3935)}{\sqrt{(100 \times 0.3935 \times 0.6065)}}\right) \\ &= P(Z \leq -1.81) \\ &= 0.0351 \end{aligned}$$

Q # 10

X has a beta distribution

$$\alpha = 1$$

$$\beta = 3$$

Probability density function of beta function

$$f(x) = \frac{x^{\alpha-1} (1-x)^{\beta-1}}{\Gamma(\alpha) \Gamma(\beta) / \Gamma(\alpha+\beta)}$$

$$0 < x < 1$$

Replace α with 1 and β with 3

$$f(x) = \frac{x^{1-1}(1-x)^{3-1}}{\Gamma(1)\Gamma(3)/\Gamma(1+3)}$$

$$= \frac{(1-x)^2}{1/3}$$

$$= 3(1-x)^2$$

(a) The mean μ is the sum of product of each possibility x with its probability $f(x)$

$$\mu = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_0^1 x^3 (1-x)^2 dx$$

$$= \int_0^1 3x - 6x^2 + 3x^3 dx$$

$$= \left(\frac{3}{2} x^2 - 2x^3 + \frac{3}{4} x^4 \right) \Big|_0^1$$

$$= \frac{3}{2} - 2 + \frac{3}{4}$$

$$= \frac{1}{4}$$

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$$\mu = \frac{\alpha}{\alpha + \beta}$$

$$= \frac{1}{1+3}$$

$$\mu = 1/4$$

The median is the value of x for which the probability to the left of x is equal to 0.5 (50%)

$$P(X < x) = \int_0^x 3(1-x)^2 dx$$

$$= \int_0^x 3 - 6x + 3x^2 dx$$

$$= (3x - 3x^2 + x^3) \Big|_0^x$$

$$= 3x - 3x^2 + x^3$$

the probability has to be equal to 0.5

$$3x - 3x^2 + x^3 = 0.5$$

this equation only has one real root

$$x = 0.206299$$

(b) Formula for the variance of a beta distribution.

$$\begin{aligned}\sigma^2 &= \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} \\ \alpha &= 1 \quad \text{and} \quad \beta = 3 \\ \sigma^2 &= \frac{1 \cdot 3}{(1+3)^2(1+3+1)} \\ &= \frac{3}{16(5)} \\ &= \frac{3}{80} \\ &= 0.0375\end{aligned}$$

(c) The probability is the integral of a probability density function over the corresponding interval

$$\begin{aligned}P(x > 1/3) &= \int_{1/3}^1 f(x) dx \\ &= \int_{1/3}^1 3(1-x)^2 dx \\ &= \int_{1/3}^1 (3-6x+3x^2) dx \\ &= (3x - 3x^2 + x^3) \Big|_{1/3}^1 \\ &= 3 - 3 + 1 - \left(1 - \frac{1}{3} + \frac{1}{27}\right) \\ &= \frac{8}{27} \approx 0.2963 = 29.63\%.\end{aligned}$$

Q # 11

Let the random variable x represents the survival time x follows a gamma-distribution with $\alpha=5$ and $\beta=10$

- (a) Let's calculate the mean survival time of the randomly selected animal of the type used in the experiment.

$$\begin{aligned} E(X) &= \mu \\ &= \alpha\beta \\ &= (5)(10) \\ E(X) &= (50) \end{aligned}$$

- (b) Let's determine the standard deviation of survival time

$$\begin{aligned} \sigma &= \sqrt{\alpha\beta^2} \\ \sigma &= \sqrt{5 \times 10^2} \\ \sigma &= \sqrt{5 \times 100} \\ \sigma &= 22.36 \end{aligned}$$

- (c) Let's calculate the probability that an animal survives more than 30 weeks

$$\begin{aligned} P(X \geq 30) &= 1 - P(X \leq 30) \\ &= 1 - \frac{1}{\beta^\alpha} \int_0^{30} \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)} dx \end{aligned}$$

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using $y = \frac{x}{\beta}$, we get

$$P(X \geq 30) = 1 - \int_0^3 \frac{y^4 e^{-y}}{\Gamma(5)} dy$$

$$= 1 - F(3; 5)$$

$$= 1 - 0.1850$$

$$= 0.815.$$